

MUHAMMAD WALEED AHMAD

AI Engineer | Computer Vision | Generative & Agentic AI

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PROFESSIONAL SUMMARY

Results-driven AI Engineer with 1+ year of hands-on experience building and deploying production-grade Computer Vision, Generative AI, and Agentic AI systems. Proven track record delivering real-world products — including a live multi-site surveillance platform (Fortix AI) and a psychology AI assistant (Sprixle) — using PyTorch, YOLOv11/v26, LangChain, LangGraph, and RAG architectures. Skilled in end-to-end MLOps: data pipelines, LLM fine-tuning, vector databases, multi-agent workflows, model deployment with FastAPI and Docker on AWS. Passionate about solving high-impact problems with intelligent automation.

TECHNICAL SKILLS

AI / ML: Machine Learning, Deep Learning, Computer Vision, NLP, Transformers, Vision Transformers, LLM Fine-Tuning, Prompt Engineering, Generative AI, Agentic AI, Multimodal AI, RAG, MLOps, Model Deployment, Real-Time Inference, Data Annotator, 3D Slicer, CVAT

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Ultralytics YOLO (v11/v26/YOLOE), OpenCV, LangChain, LangGraph, Hugging Face Transformers, Keras, FastAPI, NumPy, Pandas, N8N

DevOps & Cloud: Docker, AWS, GitHub, CI/CD Pipelines (GitHub Actions), Kubernetes

Languages: Python, JavaScript, Node.js, PostgreSQL, SQLite

WORK EXPERIENCE

Shayan Solutions | Lahore, Pakistan

Sep 2025 – Present

AI Engineer

- Computer Vision Systems:** Designed, trained, and optimized real-time object detection and tracking pipelines using YOLOv11/YOLOv26/SAM3 with ByteTrack, achieving >92% mAP on custom surveillance datasets across 6 detection categories (weapons, fire, PPE, shoplifting, vehicles, faces).
- Agentic AI Pipelines:** Built end-to-end multi-agent workflows using LangChain and LangGraph with RAG architectures and vector databases, reducing average query response latency by 40% through optimized retrieval strategies and context window management.
- LLM Fine-Tuning & MLOps:** Continuously fine-tuned LLMs using Hugging Face on domain-specific datasets; deployed models via FastAPI and Docker on AWS with GitHub Actions CI/CD, maintaining 99.5% uptime across production systems.
- Scalability & Performance:** Engineered feature engineering and data preprocessing pipelines handling 50,000+ video frames per day, improving model inference throughput by 35% through batch optimization and TensorRT integration.

Fiverivers Technologies | Lahore, Pakistan

Jun 2025 – Sep 2025

Data Scientist Intern

- ML Pipelines:** Developed end-to-end ML pipelines for 3 client classification and regression projects — covering data collection, cleaning, feature selection, and model evaluation with scikit-learn — improving F1 scores by an average of 18% over baseline models.
- LLM Fine-Tuning & NLP:** Fine-tuned 2 LLMs and 1 VLM using Hugging Face on Google Colab and Kaggle (GPU environments); built RAG-based NLP systems with LangChain and LangGraph deployed for real-world client use, processing 10,000+ documents.

PROJECTS

Fortix AI — Real-Time Surveillance System <https://www.fortixai.io/>

- Stack:** YOLOv11/v26/YOLOE, ByteTrack, ANPR, FastAPI, PostgreSQL, AWS
- Built a production web-based multi-site camera monitoring platform serving real clients; implemented real-time detection for 8 event types (fighting, weapons, fire, smoke, PPE violations, shoplifting, robbery, vehicle violations) with <200ms alert latency.

- Integrated facial recognition and Automatic Number Plate Recognition (ANPR) to automatically extract and log license plate text, reducing manual incident reporting time by ~60%.

Sprixle — Psychology Agentic AI System <https://www.sprixle.com/>

- **Stack:** LangChain, LangGraph, LLMs, FastAPI, Vector Database
- Architected autonomous AI workflows with persistent memory management and emotion/context understanding for personalized psychological assistance; system supports 500+ concurrent adaptive conversations.
- Developed scalable backend APIs enabling real-time contextual AI interactions with sub-300ms response times, deployed and live in production.

3D Room Reconstruction (Droid-SLAM + Spatial LM)

- **Stack:** Droid-SLAM, Spatial Language Model, Python, 3D Point Cloud Processing
- Implemented Droid-SLAM to generate precise 3D point clouds from monocular room images, achieving <2cm reconstruction error on test scenes; used a spatial LM to reconstruct fully editable 3D layouts for home interior planning use cases.

DeepTutor AI — Intelligent Deep Learning Chatbot

- **Stack:** unsloth/gemma3n 2B, RAG, LangGraph, Multimodal (video + audio)
- Built a domain-specific chatbot using gemma3n 2B with RAG for deep learning Q&A; integrated LangGraph for multi-step reasoning and added multimodal input (facial expression + tone analysis from video/audio), improving contextual answer relevance by 28% vs. base model.

EDUCATION

University of Central Punjab

Oct 2021 – Jul 2025

Bachelor of Science in Computer Science | Lahore, Pakistan

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Structures & Algorithms, Database Systems, Software Engineering, Artificial Intelligence

ACHIEVEMENTS & HIGHLIGHTS

- Delivered 2 live production AI products (Fortix AI, Sprixle) within first year of professional career.
- Worked across full AI stack: data collection → model training → fine-tuning → deployment → monitoring.
- Proficient in both research-oriented (custom model architecture, SLAM) and product-oriented (APIs, MLOps, CI/CD) AI engineering.